

Heterogeneous Effects of Regulatory Enforcement: Evidence from OSHA

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Abstract

Regulatory burdens are said to stifle economic activity — but what happens when regulations are enforced on noncompliant firms? I estimate how regulatory enforcement affects individual plants in the context of the US Occupational Safety & Health Administration, linking data on surprise inspections to restricted-use Census micro data. Using a matched-stacked event study design, I find that while high regulatory fines decrease employment and the probability that a plant remains open, low fines have the opposite effect. I interpret these results in a conceptual framework which emphasizes differences in profitability across workplace hazards. I consider both direct and indirect costs and benefits of remediation, finding that indirect effects likely play a greater role. High-fined plants face reputational damage, including paying a 4.5% premium to newly hired workers. Low-fined plants in contrast likely experience positive outcomes driven by spillover benefits beyond safety adaptations. These findings show that regulation’s effects are more complicated than critics allege and in some cases may even improve performance.

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I Introduction

Opponents of business regulation argue it is burdensome, costing workers jobs and stifling economic activity ([US Chamber of Commerce 2025](#); [Executive Order 14192 2025](#)). Higher regulatory burdens are associated with lower firm employment and profitability ([Branstetter et al. 2013](#); [Alfaro-Urena et al. 2022](#)), and some firms choose not to comply with regulations, preferring to risk fines than bear regulatory burdens ([Stansbury 2021](#)). If the government intervenes at these firms to enforce regulations, this argument would predict decreases in employment and profits — yet little evidence exists on how regulatory enforcement affects noncompliant plants.

In this paper I provide new evidence on the effects of regulatory enforcement in the context of the US Occupational Safety & Health Administration (OSHA). I write a stylized model of plant safety choice which allows a regulator to enforce compliance, affecting plant labor costs. I then test the predictions of this model by linking data on surprise OSHA inspections to restricted-use microdata from the US Census, estimating the effect of OSHA’s safety enforcement on plant employment, survival, and worker earnings.

My results show differing effects of enforcement depending on the size of the associated fines. Plants which receive low fines from OSHA (below the average of \$100 per worker) see an increase in employment of around 5% relative to unfined plants. Similarly, low-fined plants experience a decrease in the probability of closure of roughly 0.5 percentage points (a 50% decline), suggesting a meaningful difference in profitability post-inspection. In contrast, plants receiving high fines (over \$100 per worker) see a roughly 1% decrease in employment, and a sharp 70% increase in closure probability.

To ensure these results are causal, I use propensity score matching among plants inspected by OSHA to pair fined plants with highly-similar unfined ones. I estimate a stacked event study design to control for unobservable plant characteristics and trends across states and industries ([Lee and Taylor 2019](#); [Schmieder et al. 2023](#)). The resulting estimates are causal under the identification assumption: that the conditional difference between fined and unfined plants would have been stable in the absence of fines. I defend this assumption through pre-trend analysis and contextual details from OSHA’s inspection process which minimizes the differences among inspected plants.

To understand what drives these heterogeneous effects, consider two examples that highlight the differences across workplace hazards. Two chemicals commonly used in construction are lead and methylene chloride (MC). Lead exposure causes serious diseases, but can be safely mitigated if clothing is properly laundered. In contrast, exposure to MC has serious, but quite rare, consequences, and is costly to mitigate — requiring air filtration systems and monitoring. The resulting cost-benefit comparison for lead remediation is quite favorable to

firms, whereas the comparison for MC is unfavorable, possibly resulting in differential effects ([Occupational Safety and Health Administration 2010, 2025a](#)).

Expanding on this insight, I write a conceptual framework that generalizes plant safety choices over heterogeneous hazards. In this model, profit-maximizing plants choose to remediate workplace hazards when the benefit of doing so exceeds the costs. Plants are then visited by a surprise regulator, who enforces remediation of hazards. Plants must adjust their safety accordingly, resulting in changes in their economic outcomes.

I consider the effects of enforcement on both direct and indirect costs and benefits of remediation. While the direct components appear to be more profitable for low fines — contradicting the estimated effects on employment and survival — indirect components seem to be more aligned. High fines are associated with productivity losses in related settings ([Li 2022](#)) and I find evidence of reputational damage in the labor market: I estimate that high-fine plants face a 4.5% pay premium for new hires. I show that this effect is not explained by differences in worker talent, estimating a worker effects model following [Abowd et al. \(1999\)](#). This suggests that high fines cause indirect costs which hurt the profitability of plants. In contrast, low fines are associated with spillovers in prior literature ([Haviland et al. 2010](#); [Berman and Bui 2001](#)) which may generate indirect benefits and increase profitability ([Bloom et al. 2025](#)).

I also examine the effects of regulatory enforcement on average worker earnings¹. The theory of compensating differentials ([Rosen 1974, 1986](#)) provides testable predictions for this outcome, predicting that improved worker safety should reduce compensation. In contrast with this theory, I detect no effect of safety enforcement on earnings. This result holds for workers at both low- and high-fine plants.

The absence of wage adjustments is consistent with other prior work. The empirical evidence for adjustments to safety conditions is concentrated in findings on highly salient changes in mortality risk ([Lavetti 2020](#); [Lee and Taylor 2019](#)). Safety improvements which are general or abstract are insufficiently salient to change worker beliefs. Another explanation could be that plants rarely respond to shocks through adjustment along the wage margin ([Shimer 2005](#)). Wages may be too downward-rigid to adjust to safety changes when opposed by frictions like implicit contracts ([Beaudry and DiNardo 1991](#)) and menu costs. I contribute to this literature by identifying where and for what workers we should most expect to find compensating effects.

I also contribute to the literature on the effects of regulation through analysis of new outcomes at the plant level. Most research on regulatory effects examines firms or markets, but the effects on plants are less understood and could contain important heterogeneity. At large

¹To my knowledge, the first exploration of enforcement on this outcome in a quasi-experimental setting.

public firms, regulatory fines have direct effects, decreasing market values, though usually commensurately to the size of the fine (Karpoff et al. 2005). In county-level markets, the effects of regulation are more complicated, as greater stringency is associated with increased firm exit and weakened competition (Ryan 2012; Curtis 2020), suggesting effects beyond simply the financial burden of penalties.

Effects are more nuanced in the small literature at the plant level, where enforcement has been shown to lower key characteristics like productivity (Greenstone et al. 2012; Li 2022). Enforcement can even spillover into plant behavior in areas unrelated to the target of the enforcement (Haviland et al. 2010; Bloom et al. 2025). My paper contributes to this literature by extending plant-level analyses, specifically considering the effects of enforcement on non-compliant firms and also how effects vary with the size of the related penalty.

This paper also contributes to the small literature concerning OSHA. OSHA inspections themselves (independent of fines or enforcement) have a strong positive effect on worker safety outcomes (Levine et al. 2012; Lee and Taylor 2019). Furthermore, inspections which result in higher fines are associated with even greater improvements in safety (Haviland et al. 2010), identifying the greater effect of enforcement over inspection alone. Inspections alone have been shown to affect other economic outcomes, but the effect of fines remains only partially explored Haviland et al. (2010). This paper emphasizes the role of fines in enforcement, exploring new outcomes, and improves our understanding of how enforcement of different OSHA standards can have differential effects.

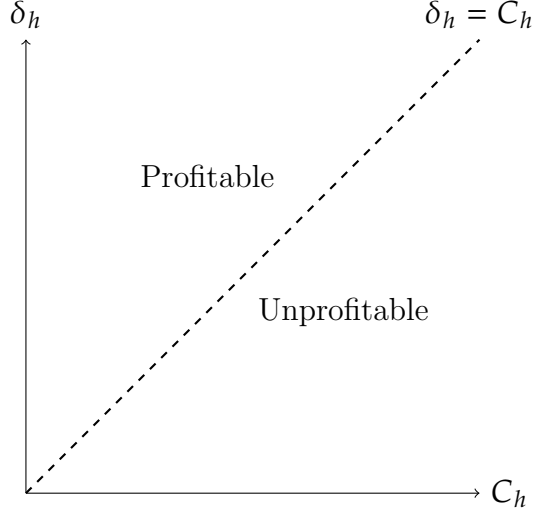
II Conceptual Framework

Consider a population of identical plants facing the same set of workplace hazards, indexed by $h \in H$. Plants can choose to remediate a hazard h , and in doing so they incur both a cost of C_h per worker (reflecting the cost of additional equipment or trainings) and benefit δ_h per worker (reflecting the reduced absenteeism and worker compensation costs that arise when the hazard results in injury). Assume $C_h \geq 0$ and $\delta_h \geq 0$.

Profit-maximizing plants will choose which hazards to remediate by comparing their costs and benefits. Under a strong assumption of perfect information (to be relaxed shortly), managers would know the value of C_h and δ_h for all h . They would then make all remediations which are profitable, i.e., those for hazards where it holds that $C_h \leq \delta_h$. This division is demonstrated in Figure 1. I realistically assume that there are some hazards for which this condition does not hold, as some safety measures are highly costly relative to their benefit — if all were clearly profitable, there would be much less need for safety regulation.

Consider next a regulatory inspection, where a regulator enforces remediation of some hazard h^* to comply with safety regulations. For h^* to be unremediated when the inspection

Figure 1: Hazard Remediation Space



Note: The x-axis measures the cost of remediating a hazard, and the y-axis the benefit, with the 45°-line demarcating the boundary between profitable remediations (above) and unprofitable ones (below).

occurs, it must be the case that $C_{h^*} > \delta_{h^*}$, or else it would have been profitable to remediate *ex ante*. This condition means that remediation of h^* results in an effective increase in costs per worker for the plant. Under most reasonable assumptions about the market structure, this should result in reduced employment, higher prices, reduced sales, and lower profits. In turn, this unprofitability would imply decreased survival for the plant affected.

Relaxing the assumption of perfect information, consider a case where plants do not perfectly perceive the cost or benefits of hazards but have beliefs about them². Given OSHA's enforcement of thousands of individual standards relating to thousands of potential hazards, this relaxation seems warranted, especially given managers with realistically limited scopes of control. Under this setting, plants choose *ex-ante* to make all remediations which they *perceive* to be profitable, where their belief in the hazards cost $\tilde{C}_h$ is less than their belief in the benefit $\tilde{\delta}_h$. I abstract from the belief-generating process, but it could be driven by factors such as inertia in safety choices.

When OSHA now inspects such a plant, and mandates remediation of an unaddressed hazard, it may be that the hazard is profitable or unprofitable for the plant, depending on the plant's beliefs. Accordingly, it could be the case that OSHA-enforced remediation has the same effect as in the perfect information case (enforcing an unprofitable remediation) or it could be that it has the opposite effect (enforcing a profitable remediation). If OSHA mandates a profitable remediation, the plant would spend C_h and achieve a benefit

²This modification is to introducing frictions that generate sub-optimal behavior by firms like the managerial inertia described in [DellaVigna and Gentzkow \(2019\)](#).

of $\delta_h > C_h$ per worker. This effective decrease in the cost of labor should result in increased employment, lower prices, increased sales, and increased profits. Plants in such a case would accordingly see increased survival rates relative to the unaffected.

To ground the contrast between potentially profitable and unprofitable hazards in reality, consider the example of exposure to two chemicals commonly found on construction sites: lead and methylene chloride (MC). Protecting workers from lead poisoning rarely requires any special equipment, and usually it suffices that any contaminated clothing be cleaned and laundered before reuse ([Occupational Safety and Health Administration 2024a](#)). Protection from methylene chloride on the other hand requires investment in respiratory and air filtration systems (estimated to cost roughly \$4,000-\$9,000 per unit), as well as ongoing costs of monitoring and medical evaluations ([Occupational Safety and Health Administration 2024b](#); [Occupational Safety and Health Administration 2010](#)). Clearly, the cost of remediation on a per-worker basis will be much lower for lead than MC.

A similar case holds for the benefits to the firm. Lead exposure is strongly connected to a variety of serious ailments, as well as affecting cognition and worker productivity or possibly even the families of workers through secondhand exposure ([Occupational Safety and Health Administration 2025a](#)). Preventing this helps the firm avoid significant costs in worker absence and medical expenses (via worker's compensation claims), while also avoiding lost productivity through cognition declines. In contrast, OSHA's own estimates for the benefit of full compliance with methylene chloride remediation nationwide is only 34 saved lives per year ([Occupational Safety and Health Administration 2010](#)). While any such death is tragic, the probability of exposure-induced outcomes is simply so rare that the expected benefit of methylene chloride remediation for any one plant is low.

These two chemicals provide just one example of the variation that exists in costs, benefits, and the balance between the two across hazards. Note that this variation can also be viewed in terms of social cost-benefit versus firm cost-benefit. OSHA as a public regulator seeks to avoid requiring standards which are socially inefficient, where the costs to firms exceed the combined benefits to workers and firms; however, standards may be socially efficient and still unprofitable to the firms. This is most likely the case when there are benefits which accrue to workers (or the public healthcare system) which, when combined with the benefits to the plant, exceed the costs. The firm bears all of the costs, but the benefits are shared among the firm, workers, and society more broadly.

Imperfect information over costs and benefits, whether derived from an imperfect belief-formation process or frictions like managerial inertia, demonstrates how different effects of regulatory inspections across plants could arise. While additional extensions are possible (including uncertain inspections and risk aversion, explored in [Appendix A](#)) this basic model provides sufficient framework for interpreting the empirical results.

III Data

III.A Background on OSHA: Inspections & Fines

OSHA provides a rich context in which to study the effects of regulatory enforcement. OSHA is a United States federal regulatory agency with the mission to “assure America’s workers have safe and healthful working conditions”. It both sets safety standards and enforces them, identifying the methods by which firms must maintain safety and regularly inspecting workplaces for compliance with those methods. There are nearly 1,000 individual OSHA standards, and any given firm may be subject to several hundred at once.

OSHA enforces standards primarily through inspections of workplaces. To organize inspections, OSHA divides the country into 10 broad regions and then further into 85 Areas, each with its own office³. Each Area Office selects the plants in their jurisdiction for inspection, with two broad categories of selection criteria. “Programmed” inspections are targeted at specific industries or hazards, as identified by either National or Local Emphasis Programs, which consider priorities for enforcement at each level. Among the plants which meet the criteria for a given Program, OSHA usually then randomly assigns inspections to a subset of eligible plants, given their limited capacity. The second category of inspection is those which are initiated by some event. These could be formal complaints (lodged by a worker or concerned citizen), referrals (lodged by another Area Office, agency, or media organization), or major accidents (such as a worker fatality). While OSHA completes tens of thousands of inspections annually, their capacity limit typically means that they cover less than one percent of the over 8 million workplaces under their jurisdiction⁴.

During an inspection, inspectors tour the facility (or broader work site in cases like construction) and review employer records to assess compliance with standards ([Occupational Safety and Health Administration 2025b](#)). This inspection is done alongside representatives of the firm, though normally without any advance notice. If a facility is found to be out of compliance with a standard, the inspector issues a citation for each separate violation. The inspection concludes with a conference between the employer representative and the inspector to discuss the results, after which the inspector returns to their Area Office to determine the relevant financial penalty or fine.

Fines themselves are calculated in accordance with a detailed process set out in the Occupational Safety and Health Act of 1970. This bases fines primarily in the gravity of the violation, which itself is determined by the combination of two sub-factors: the severity

³OSHA has direct jurisdiction over 28 states; the other 22 operate federally-approved state-run safety and health programs.

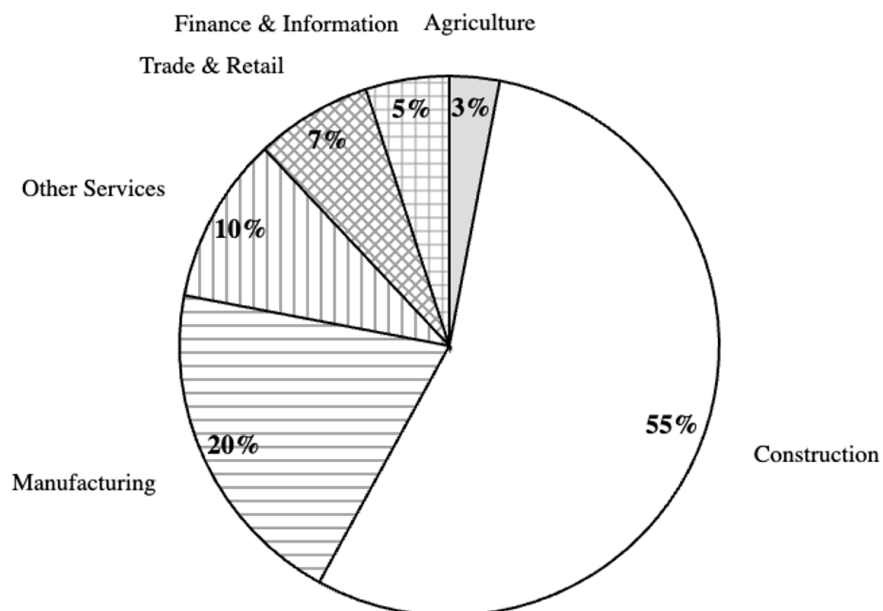
⁴US Department of Labor, Occupational Health and Safety Administration, “Commonly Used Statistics,” <https://www.osha.gov/oshstats/commonstats.html>, accessed October 2025.

of the injury which could result from the violation and the probability that such an injury could occur. The severity is assessed across categories ranging from de minimus to high severity (i.e., death or permanent disability), while the probability takes into consideration the number of employees exposed, the frequency and proximity of the hazardous condition, and relevant trainings available. Fines are adjusted further to reflect the size of the employers business, as well as their good faith / history of previous violations. Once fines are levied, OSHA follows-up with plants to ensure remediation in a timely manner.

III.B OSHA Inspection Data

I use public data on OSHA inspections from the US Department of Labor Data Catalog. These provide detailed information about each inspection that OSHA conducts, including inspection type (Programmed, Initiated), plant characteristics (industry, union status, name and address), and outcomes of the inspection (violations, fines, employees affected, statutes violated).

Figure 2: Industry Shares of Inspections



Note: The share of all plants inspected by OSHA in the sample that belong to each one-digit NAICS industry code are shown.

I restrict my sample of inspections in several ways in order to cleanly identify most the effect of enforcement. To focus my analysis on inspections which are most plausibly unanticipated by the plant, I use only Programmed inspections and remove from the sample all inspections which occurred for reasons of Complaints, Referrals, or Accidents. Additionally, I use the full history of inspections to identify those plants which are experiencing their first

inspection, and use only those. Finally, I restrict the sample to inspections occurring between 2000 and 2015, so that I can observe each plant for four years before and after inspection. These year cutoffs are chosen because the Census data begins in 1996 and because I want to avoid data for 2020-onwards, when the Covid-19 pandemic significantly affected OSHA operations and other economic outcomes.

Figure 2 shows a breakdown of inspections by industry. As expected, the highest shares are in those industries which include the most routine exposure to hazards, Construction and Manufacturing, accounting for three quarters of all inspections. The remaining industries are less represented, reflecting the strongly targeted nature of OSHA inspections⁵

Motivated by the conceptual framework, I analyze my sample of inspections to calculate the share of each standard across all violations, as well as the average fine-per-worker for each standard. I provide six examples below in Table 1: three associated with relatively low fines and three with relatively high fines, all of which are among the most common standards violated. These examples suggest that there is significant heterogeneity across hazards / standards.

Table 1: Example Common OSHA Violations

Violation	Share of All Violations	Average Fine / Worker
Portable Fire Extinguishers	0.5%	\$50
Labeling of Hazards	0.5%	\$60
Providing PPE	1%	\$80
Machine Guards & Training	1%	\$2,700
Eye & Face Protection	2%	\$1,500
Fall Protection System	4.5%	\$2,400

III.C Census Plant Data

To analyze long-term outcomes I link to the Longitudinal Business Database (LBD) at the US Census. The LBD is a rich annual dataset of plant-level information, covering almost all private establishments. It contains detailed firm variables, including several which are key to my analysis such as the headcount and payroll of the plant, as well as the age of the plant’s broader firm. Additionally, because the LBD tracks individual establishments over time, the closure or “death” year of a plant is identified. I use data from 1996 to 2019, allowing each plant to be observed up to four years before and after inspection. Linking the OSHA data to the LBD relies on the presence of name and street address information in both datasets.

⁵Mining is omitted from this analysis because of the separate agency (the Mine Safety & Health Administration) which administers safety in that industry.

This linkage is a multi-step process, described in more detail in [Appendix B](#). I follow best practices common in Census linking as enumerated in [Chow et al. \(2021\)](#).

Some outcomes of interest are not available in the LBD itself, so I additionally link from the LBD to the the Longitudinal Employer-Household Dynamics (LEHD) Quarterly Workforce Indicators (QWI) dataset. This data uses administrative employer-employee histories to generate plant-level variables at the quarterly level, including hirings, separations, and earnings of both incumbent and newly hired full-time workers. I aggregate these variables to annual averages for consistency with the frequency of the LBD. The final sample includes approximately 16,400 plants⁶.

III.D Propensity Score Matching

Comparisons of fined and unfined plants may suffer potential differences between these groups which correlate with both misbehavior and outcomes of interest. While many of these differences can be econometrically remedied through plant-level fixed effects and other controls, it is more difficult to control for differential trends between the groups. To overcome this imbalance, I use propensity score matching to obtain a comparison set of unfined plants which are highly similar to the fined set in both levels and trends of relevant variables. I match plants from the same state, industry, and year of inspection on the propensity to be treated given the value of key variables in the year prior to inspection. These matching variables include firm age, firm size, plant size, local HHI, plant payroll, plant average earnings, union status. For each treated fined plant, I assign a single unfined control within its state-year-industry cell, using the plant with the closest propensity score (without replacement). Within each cell, the closest matches are prioritized for producing matched pairs.

This matching yields a final analysis sample of fined plants and highly similar unfined plants evolving on similar trends pre-OSHA inspection, described in [Table 7](#). This approach is similar to the suggestion of [Abadie \(2005\)](#) to use propensity score reweighting to create a control group with similar pre-treatment characteristics for which the parallel trends assumption is more viable. Creating an explicit control group in each state-year-industry cell and subsequently stacking the cells into one dataset also has the advantage of avoiding the complications of staggered event study timings arising from multiple times of treatment as described in [Goodman-Bacon \(2021\)](#). This approach of stacking matched treatment-control groups for every treatment period is similar to the related estimator proposed by [Callaway and Sant’Anna \(2021\)](#) for this econometric situation, as recently applied in, e.g., [Schmieder et al. \(2023\)](#).

⁶The Census requires rounding of key statistics such as sample sizes.

Table 2: Summary Statistics

	Full Sample		Matched Sample	
	Control	Treated	Control	Treated
Firm Age	20.33 (11.850)	19.40 (11.510)	20.66 (11.500)	20.10 (11.530)
Log Plant Size	3.11 (1.540)	3.15 (1.453)	3.15 (1.510)	3.16 (1.420)
Log Firm Size	4.25 (2.956)	3.86 (2.544)	4.20 (2.857)	3.99 (2.611)
Log Quarterly Worker Earnings	8.01 (0.587)	7.95 (0.538)	8.02 (0.570)	7.97 (0.516)
Log Payroll (000s)	6.62 (1.747)	6.59 (1.678)	6.69 (1.707)	6.64 (1.596)
Log Quarterly New Hire Earnings	7.61 (0.696)	7.57 (0.649)	7.63 (0.686)	7.57 (0.642)
County-Sector Employment HHI	0.306 (1.402)	0.280 (0.876)	0.270 (0.689)	0.270 (0.624)
Plants	5,900	10,500	3,600	3,600

Note: Means are provided with standard deviations in parentheses.

IV Empirical Approach

My main econometric estimates are derived from the following event study specification:

$$y_{jtb} = \theta_j + \gamma_{st} + \alpha_{it} + \beta X_{jt} + \sum_{\tau=-4}^4 \beta_{\tau} \mathbb{1}\{t = b + \tau\} + \sum_{\tau=-4}^4 \delta_{\tau} \mathbb{1}\{t = b + \tau\} \text{Fine}_j + \epsilon_{jtb} \quad (1)$$

where y_{jtb} is the outcome of interest y for plant j in year t , and the plant was inspected by OSHA in baseline year b . I include fixed effects for the plant, θ_j , the state-year, γ_{st} and industry-year, α_{it} . X_{jt} contains time-varying plant controls, specifically firm age and HHI of local sectoral employment⁷. Finally, two sets of event study coefficients are estimated. Each set omits the period of $\tau = -1$, the year before inspection, as the reference year. The

⁷The importance of labor market power in workplace conditions is shown in work like [Lavetti and Schmutte \(2024\)](#).

first set, β_τ , accounts for changes relative to inspection affecting all plants. The second set, δ_τ , are the coefficients of interest, the estimated difference between fined and unfined plants. The variable Fine_p is an indicator variable for whether plant p was fined as a result of their inspection. Finally, ϵ_{jtb} is the error.

For every outcome of interest I also estimate the corresponding difference-in-difference model, with related specification:

$$y_{jtb} = \theta_j + \gamma_{st} + \alpha_{it} + \beta X_{jt} + \beta \mathbb{1}\{t \geq b\} + \delta \mathbb{1}\{t \geq b\} \text{Fine}_j + \epsilon_{jtb} \quad (2)$$

where the event study coefficients are simplified to β capturing the differences between pre- and post-inspection, and δ capturing the difference in the difference between fined and unfined plants pre- and post-inspection.

The key assumption for both of these models to identify the causal effect of enforcement is the conditional parallel trends assumption. This assumption means that in the absence of OSHA enforcement, the difference between the fined and unfined plants would not have changed over time. While not directly testable, the absence of differential pre-trends in the results shown in [Section V](#) support this assumption. Furthermore, timing is effectively random conditional on industry-state-year, preventing anticipation effects.

IV.A Heterogeneity by Fine Size

The conceptual framework predicts that enforcement of different OSHA standards may have different effects on plants, leading me to incorporate heterogeneity analysis. While grouping text-based standards is complex, a more natural extension is to consider the effects of variation in the size of the fine / enforcement penalty applied to plants. It is intuitive to believe that fines of different sizes should have different effects on plants — in the extreme, it would be shocking if a fine of only a few hundred dollars was considered comparable to one in the millions.

To account for this potential heterogeneity, I conduct heterogeneity analysis by separately estimating these specifications over two sub-samples: a low-fine sample and a high-fine sample⁸. To distinguish between low and high fines, I use a threshold of \$100 per worker in the fine. This roughly splits the sample into the bottom two-thirds and the top third, and is near the mean across all inspections (reflecting the upward-skew of fines generally). While ideally I would test multiple thresholds for robustness, the Census places strict limitations on the

⁸The balance between treated and control units holds within each sub-sample, as detailed in [Appendix C](#).

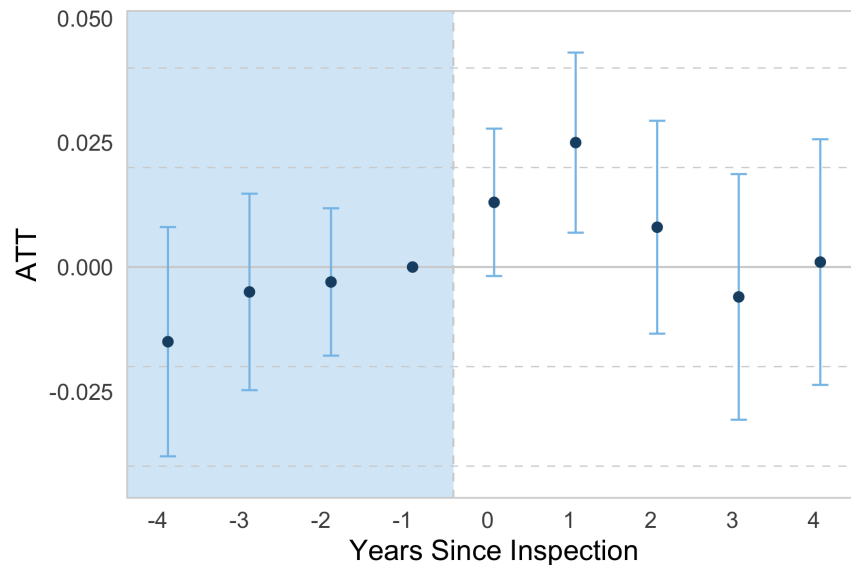
number of estimates which are disclosed for each project, making release of such robustness infeasible.

V Employment & Survival Effects

I estimate Equation (1) with the outcome of log plant employment; the resulting event study coefficients are displayed in Figure 3. These results for the full sample reveal a small effect of a roughly 2% increase, which has dissipated within two years post-inspection. Applying this effect to the average plant size in this sample (40), we would expect a brief increase of about one additional worker. On its own, this result is a weak rejoinder on the broader argument that regulation depresses employment — while not a negative effect as the argument would suppose, it is only a weak (and brief) positive one.

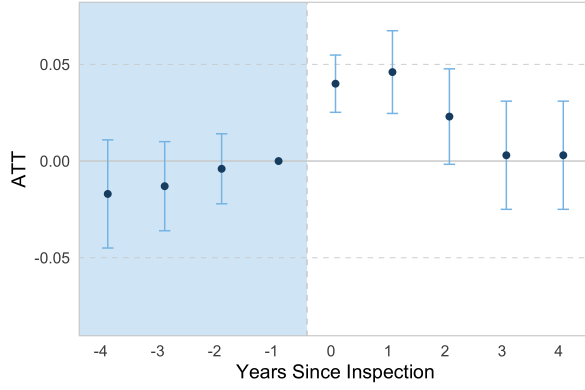
The results by size of fine describe a more nuanced effect for regulatory enforcement. As shown in Figure 4 and Figure 5, when fines are low, plants increase employment by nearly 5% at peak, while high fines lead to similar-sized decreases in employment immediately after enforcement. In both cases, the effect appears to dissipate over a few years. Interpreting these effects for the average plant, this would imply a gain (and loss) of about two workers.

Figure 3: Event Study ATT Estimates - Log Plant Employment



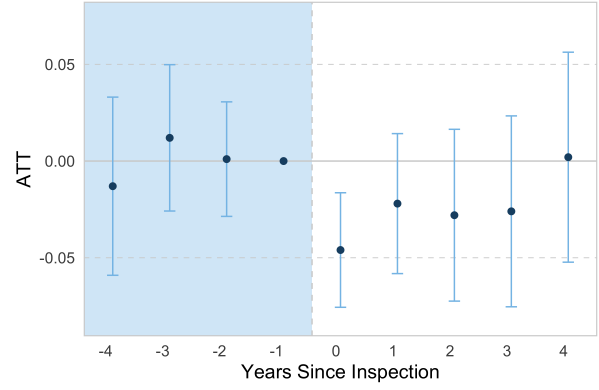
Note: Event study coefficients from Equation (1) for log plant employment relative to the inspection year (year 0), with 95% confidence intervals. The year before inspection is omitted (year -1). The shaded area represents the pre-inspection period.

Figure 4: Log Emp - Low Fine vs. None



Note: Event study coefficients Equation (1) for log plant employment in the low-fine sample with 95% confidence intervals. The year before inspection is omitted (year -1).

Figure 5: Log Emp - High Fine vs. None



Note: Event study coefficients Equation (1) for log plant employment in the high-fine sample with 95% confidence intervals. The year before inspection is omitted (year -1).

The results of the related Equation (2) difference-in-difference models for all three samples (all, low-fine, high-fine) are reported in Table 3. While the estimated coefficients for the interaction of post-inspection and fine status are aligned in direction with the broader pattern described in the event studies, the estimates are not statistically significant, possibly due to the brief duration of the effect. This duration is unlikely to be driven by a return to noncompliance. OSHA often uses prior violation record as a factor in re-inspection assignment, and the penalties for repeat violations are an order of magnitude higher than first-time offenders.

Table 3: Difference-in-Difference Effects on Employment

	Log Plant Employment		
	(1)	(2)	(3)
	All Fines	Low Fine	High Fine
Post $\times$ Fine	0.009 (0.01) [0.366]	0.017 (0.01) [0.147]	-0.009 (0.02) [0.630]
Plant FE	✓	✓	✓
State-Year FE	✓	✓	✓
Industry-Year FE	✓	✓	✓
R^2	0.923	0.944	0.944
N	53,500	37,000	16,500

Note: Standard errors are clustered at the matched-pair level in parentheses; P-levels in brackets.

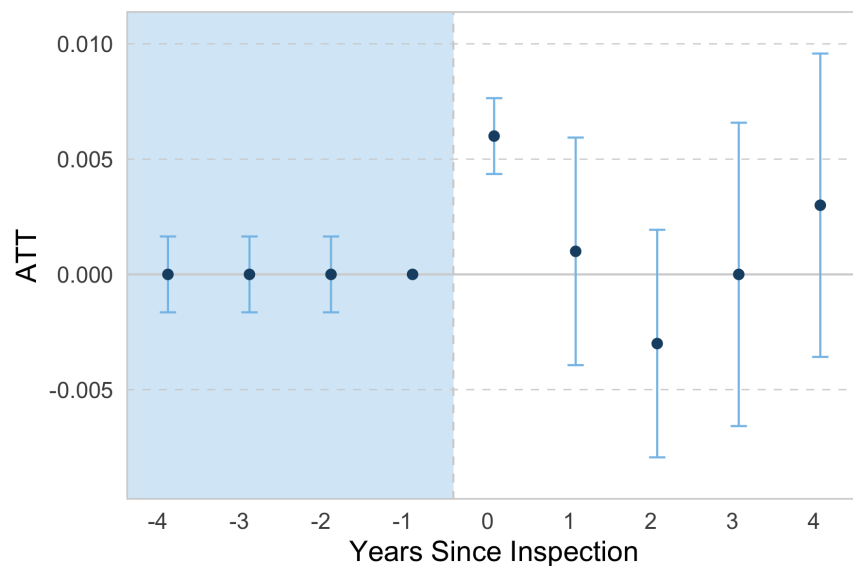
* = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

The second outcome of interest is the probability of a plant closing down or going out of business. Economists usually view this outcome as a proxy for plant profitability, as plants which cannot turn a profit are those likely to close (Curtis 2020). In the LBD dataset, this is traditionally measured as the first year in which a plant reports no employment and remains without employment for additional years.

The results here are for the marginal probability of closure, i.e., the probability that a plant which was open in the previous period closes in the period of interest. Plant closure or death is more often modeled as an absorbing state, i.e., a cumulative probability rather than marginal. Presenting the results of such a model is complicated slightly by Census disclosure rules, but the results are unlikely to differ qualitatively from those presented here.

As with the outcome of employment, the results for the full sample in Figure 6 display a small, brief effect. In this case, it appears that plants which are fined are approximately 0.5 percentage points more likely to close than those which were not fined in the year of their inspection. This small absolute magnitude is large in relative terms, as only 1% of plants close in any given year on average — making the increase a 50% jump in the likelihood of closure. The estimates for years after inspection are imprecise.

Figure 6: Event Study ATT Estimates - Plant Marginal Closure Probability

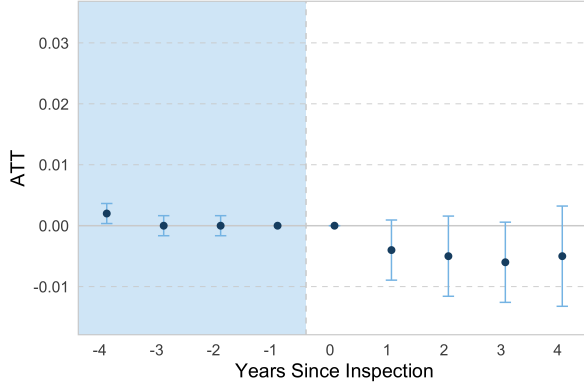


Note: Event study coefficients from Equation (1) for marginal probability of plant closure are plotted, with 95% confidence intervals. The year before inspection is omitted (year -1). The shaded area represents the pre-inspection period.

Again, the simple effect displayed in the full sample belies contrasting effects when separated by low versus high fines. As shown in Figure 7, plants which receive a relatively

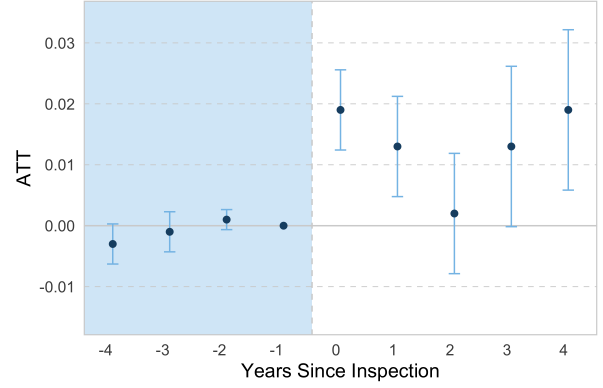
low fine are actually less likely to close after inspection than plants that receive no fine. This result persists throughout the observation period, and is also reflected in the difference-in-difference specification, with a statistically significant effect of 0.5 percentage points in [Table 4](#). In contrast, plants receiving high fines are not only more likely to close in the year of inspection, but also in the following years, up to the end of the observation period.

Figure 7: Closure - Low Fine vs. None



Note: Event study coefficients [Equation \(1\)](#) for marginal probability of plant closure in the low-fine sample with 95% confidence intervals. The year before inspection is omitted (year -1).

Figure 8: Closure - High Fine vs. None



Note: Event study coefficients [Equation \(1\)](#) for marginal probability of plant closure in the high-fine sample with 95% confidence intervals. The year before inspection is omitted (year -1).

Table 4: Difference-in-Difference Effects on Marginal Closure Probability

	Marginal Probability of Plant Closure		
	(1) All Fines	(2) Low Fine	(3) High Fine
Post $\times$ Fine	-0.001 (0.003) [0.591]	-0.005* (0.003) [0.066]	0.007 (0.005) [0.159]
Plant FE	✓	✓	✓
State-Year FE	✓	✓	✓
Industry-Year FE	✓	✓	✓
R^2	0.224	0.216	0.233
N	57,500	38,500	18,500

Note: Standard errors are clustered at the matched-pair level in parentheses; P-levels in brackets.

* = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

V.A Profitability Interpretation

A simple explanation of these diverging results based on the framework in [Section II](#) would be that plants have imperfect information about hazards. If plants only remediate those hazards that they believe to be profitable, OSHA inspections would then enforce remediation of the remaining hazards. These could be unprofitable, increasing labor costs and lowering employment and survival, but may also be profitable, lowering labor costs and increasing employment and survival.

This interpretation of the results has a direct implication for the pattern of profitability across fines: low fines would need to be associated with profitable remediations, and high fines with unprofitable ones. Assessing this relationship between fines and profitability is difficult. While it is easy to calculate the average fine for each standard from the data, profitability is complex to measure, comprising both benefits and costs, direct and indirect.

V.A.1 Direct Profitability

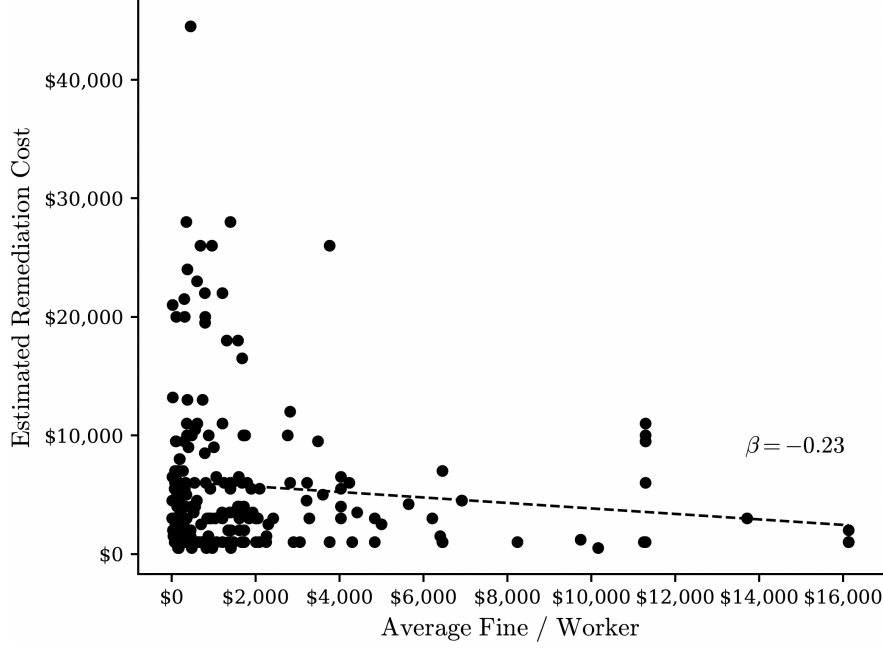
I first consider the implications of OSHA enforcement for direct profitability: the most clearly related changes in benefits (reduced costs associated with injuries) and costs (new equipment / capital and monitoring programs). For with direct benefits, OSHA’s procedural guidelines suggest that there should be a positive relationship with fines. The main factor in determining fines is the gravity of the potential harm resulting from the hazard, including the severity and probability ([Occupational Safety and Health Administration 2025b](#)). The most direct benefit of remediation to the plant comes from avoided costs of injuries (through mitigated workers’ compensation claims or absenteeism) implying that fines should be positively correlated. Of course, the pass through from harm avoided for workers to avoided costs for the firm is not complete, and may be small in cases where cause and effect are more ambiguous (e.g., long-term development of disease).

In contrast to benefits, direct costs of remediation are not officially considered in fine determination. While a comprehensive analysis of how costs vary across OSHA’s nearly 1,000 standards would be cost prohibitive to undertake, I provide directional evidence through a text analysis exercise. I scraped the text of over 200 OSHA standards from the electronic federal register, pulling those which were most readily identifiable as standalone (rather than ancillary or supplementary rules). I then processed the text using a keyword algorithm which provided estimated direct costs of implementation for each standard⁹.

The correlation between average fines and the text-based measure of direct remediation cost is depicted in [Figure 9](#). The relationship appears negative, with an OLS coefficient of \$0.23 lower estimated cost per dollar increase of fine.

⁹The algorithm is detailed in [Appendix D](#).

Figure 9: Estimated Direct Remediation Costs versus Average Fines



Note: Fines were calculated from the sample of all OSHA inspections. Remediation costs were estimated using a keyword algorithm.

This direct profitability analysis contradicts the empirical results. High fines appear associated with lower direct costs, and higher direct benefits from avoided injuries, yet these plants experience negative employment and survival effects. Low fines are associated with higher direct costs and lower direct benefits, yet these plants thrive after remediation. This contradiction implies that indirect costs and benefits, examined in the next section, must play a significant role in overall profitability of remediation; however, further study which more comprehensively assesses these variables across standards is warranted.

V.A.2 Indirect Profitability

Indirect costs of safety regulation include factors like productivity loss or reputational damage. While these are challenging to estimate based on text analysis as above, productivity costs in particular are known to be a significant component of compliance costs (Curtis 2020). This is especially true in the safety context. Studying the Mine Safety and Health Administration (MSHA), Li (2022) shows that enforced remediation of hazards causes productivity losses which far outweigh the benefits from reduced injury rates, particularly for the highest fines.

The MSHA uses a similar gravity-based procedure for determining fines, suggesting that a similar effect could be observed for OSHA fines. If productivity costs are larger for high-fine

violations — a plausible assumption given that severe hazards can require more disruptive operational changes as in [Li \(2022\)](#) — this could imply a positive relationship between fines and overall costs (direct plus indirect).

Firms are known to act in response to reputational concerns relating to regulatory compliance, including regulation of wage theft ([Ji and Weil 2010](#)) and safety in particular ([Johnson 2020](#)). More broadly, [Mas \(2008\)](#) shows that when noncompliance is associated with worse quality, there may be effects on product markets as well. While OSHA enforcement does not necessarily result in public information about violations, OSHA publishes press releases in many cases, and they are often featured in media coverage ([Johnson 2020](#)). Furthermore, violations associated with high fines are much more likely to be publicized, both due to OSHA’s own rules and public interest.

While my Census project does not allow for analysis of sales or other product market outcomes, I do have access to labor market outcomes, particularly worker earnings. I explore the effects of regulation on these outcomes separately in [Section VI](#). I find that high fines result in increased earnings costs for new-hire workers, supporting the argument that indirect costs are higher for high fines.

The indirect benefits of remediation can be conceptualized as spillover improvements from the area of remediation into other aspects of production. There is literature evidence to support the existence of such spillovers, including specifically in safety: [Haviland et al. \(2010\)](#) finds that enforcement of safety standards results in improvements not only in the specific standards which were enforced, but across other aspects of safety maintenance as well.

The act of bringing plants into compliance with regulations could provide a context for plants to reform other aspects of their operation. Enforcement in this view could act as a catalyst, generating effects beyond what *ex-ante* remediation would achieve. The text-based estimation of costs in [Section V.A.1](#) suggests that it is the low-fine, high-cost remediations that have the greatest positive effects. Higher remediation costs are often driven by capital improvements, which can increase labor demand as shown in [Berman and Bui \(2001\)](#). These larger, physical manifestations of safety changes could provide more salient opportunities for spillover effects; however, without measures of these spillovers available, this mechanism remains suggestive.

Another form of spillover could be in terms of the management quality at the plant. Interest in this factor stems from ongoing joint work which provides evidence of a strong relationship between plant safety and management. In [Bloom et al. \(2025\)](#), we use similar OSHA data in combination with surveys on management practices to show that OSHA enforcement results in improved management practices at inspected plant. Given the strong

literature on the connection between management and productivity ([Bloom and Van Reenen 2007](#); [Bloom et al. 2021](#)), this effect suggests that enforcement could have further indirect benefits which are relevant for firm profitability.

These indirect costs and benefits contrast sharply with the direct ones in [Section V.A.1](#). Productivity costs and reputational concerns are lower for low-fine violations than high-fine, while the spillover benefits from capital investment and management changes are higher. While it is not possible to compare the magnitude of the direct and indirect profitability considerations, the revealed profitability results from survival support the view that indirect costs and benefits play a significant role.

VI Worker Earnings Effects

I now investigate the effects of enforcement on labor markets through the outcome of worker earnings. While the argument that regulation burdens businesses does not directly imply an effect on wages, there is a strong existing literature in economics which makes predictions for the effect of safety regulation in particular.

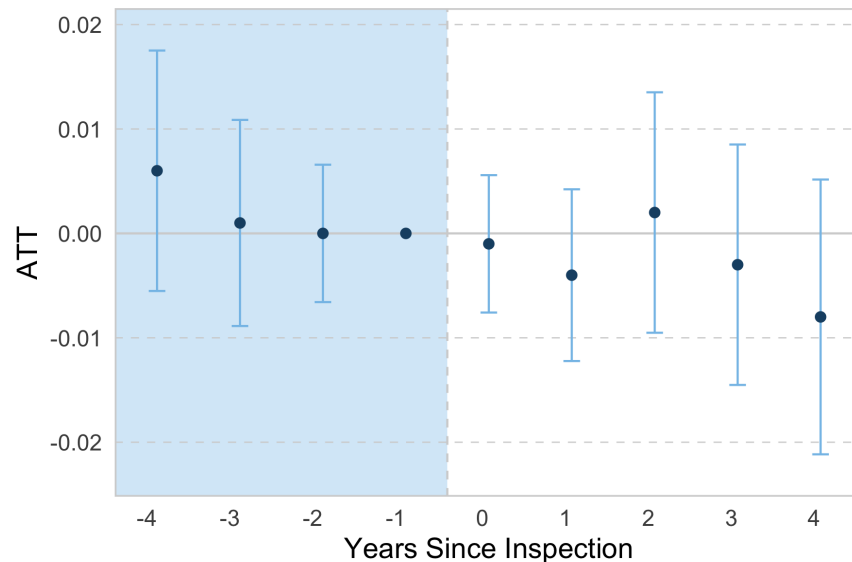
First proposed in [Rosen \(1974\)](#) and [Rosen \(1986\)](#), the theory of compensating differentials conceptualizes worker pay as reflecting not only worker contributions to production, but also their exposure to amenities. Amenities could be positive aspects of a job (e.g., free coffee) or negative ones (e.g., unsafe conditions), and affect the utility from the job accordingly. If workers are not compensated for their exposure to amenities, the resulting labor market equilibrium is inefficient: firms are paying workers more than they need to for jobs with net-positive amenities, and underpaying workers for jobs with net-negative amenities.

To apply this argument in my setting, it must be true that OSHA has positive effects on workplace safety as an amenity. While not shown in this paper, OSHA has been shown to have strong positive effects on safety elsewhere in the literature. The effect is generally established in [Levine et al. \(2012\)](#), and further quantified in mortality terms by [Lee and Taylor \(2019\)](#). More closely related to my empirical design is [Haviland et al. \(2010\)](#), who specifically show that higher fines in OSHA inspections are associated with greater improvements in safety. Accordingly, for this paper I will assume that plants which are fined experience greater improvements in safety than those which are not fined, and that this difference is greater still for the high-fine group.

In view of this existing literature, the prediction would be that worker earnings should fall in response to OSHA enforcement events. Workers who are made safer should not longer need to be compensated as much for their previously unsafe conditions, so their earnings should fall (or not rise as quickly) relative to workers whose safety is unchanged.

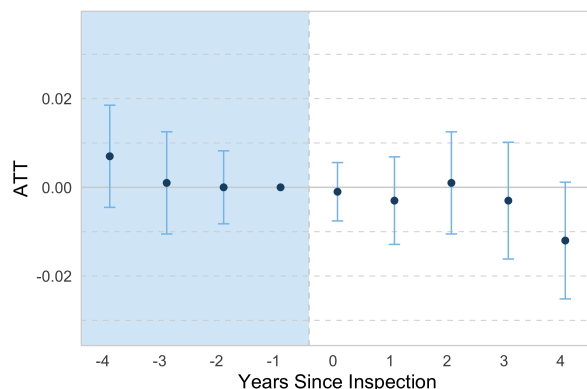
I estimate a precise null effect on earnings, presented in [Figure 10](#). The difference-in-difference version in [Table 5](#) bounds the effect between ± 1.5 percent change. Unlike in the previous two outcomes, this pattern is not overturned when analyzing the low- and high-fine groups: in both samples, no discernible effect is detected.

Figure 10: Event Study ATT Estimates - Log Average Quarterly Worker Earnings



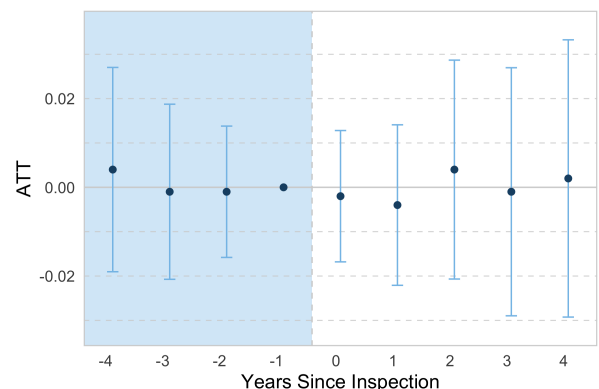
Note: Event study coefficients from [Equation \(1\)](#) for log quarterly full-time earnings to the inspection year (year 0), with 95% confidence intervals. The year before inspection is omitted (year -1). The shaded area represents the pre-inspection period.

Figure 11: ATTs - Earnings, Low Fine



Note: Event study coefficients [Equation \(1\)](#) for log quarterly full-time earnings in the low-fine sample with 95% confidence intervals. The year before inspection is omitted (year -1).

Figure 12: ATTs - Earnings, High Fine



Note: Event study coefficients [Equation \(1\)](#) for log quarterly full-time earnings in the high-fine sample with 95% confidence intervals. The year before inspection is omitted (year -1).

Interpreting this result, note that my outcome of earnings masks potential offsetting effects on the underlying factors of earnings. If we simply consider earnings as hours by wage, then it could be the case that earnings remain flat while wages fall and hours increase.

This would be aligned with the result of [Lee and Taylor \(2019\)](#), who find in a similar OSHA context a decrease in total payroll per hour worked at the plant level ¹⁰. Indeed, an increase in hours worked in response to a safety enforcement has intuitive appeal, as safer, more cautious work is likely to be slower. Unfortunately, hours are not available for analysis under the Census project of which this paper is a part.

Table 5: Difference-in-Difference Effects on Worker Earnings

	Log Average Quarterly Worker Earnings		
	(1)	(2)	(3)
	All Fines	Low Fine	High Fine
Post $\times$ Fine	0.000 (0.005) [0.470]	0.000 (0.003) [0.362]	0.000 (0.011) [0.993]
Plant FE	✓	✓	✓
State-Year FE	✓	✓	✓
Industry-Year FE	✓	✓	✓
R^2	0.902	0.912	0.885
N	53,500	37,000	16,500

Note: Standard errors are clustered at the matched-pair level in parentheses; P-levels in brackets.

* = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

While a null effect is unexpected given the literature above, there are other extant arguments in the literature which could explain this result. One possibility is that the changes in safety caused by OSHA enforcement are simply too small to be salient for worker utility. Consider the foregoing example of methylene chloride — full compliance nationwide is expected to avert only 34 deaths nationwide. For the average worker, this change in expected mortality is inconceivably low, and unlikely to therefore justify a change in earnings. This aligns with previous findings from [Viscusi and O'Connor \(1984\)](#) that workers are not fully informed about their job hazards, and may not respond to imprecise small signals.

The importance of amenity salience for compensating differentials to function is supported by the literature. Most of the evidence for compensating differentials is static — documenting systematic differences in earnings across occupations or industries which are explained by amenities. As stressed in [Lavetti \(2020\)](#)'s overview of the literature, there is scarce dynamic evidence for within-job changes in earnings which correspond to changes in amenities. The quasi-experimental evidence that does exist tends to focus precisely on

¹⁰There are other differences between our papers which prevent a direct comparison, such as differences in industry focus and sample years.

mortality risk as the most salient (Lee and Taylor 2019; Lavetti and Schmutte 2024).

A second explanation is more general: it is possible that firms are unlikely to respond to safety changes through adjustment along the wage margin due to frictions. As originally highlighted in Shimer (2005), this concept of downward nominal wage rigidity is well established in the literature (Fallick et al. 2022; Schaefer and Singleton 2023). Even if workers did recognize the benefits of greater safety, their wages could be too downward-rigid to change due to a variety of frictions, such as implicit contracts, menu costs, and general resistance to pay cuts.

VI.A New Hire Earnings

To test the strength of the downward rigidity explanation for null earnings results, I estimate the effect of enforcement on specifically newly hired workers. The earnings of newly hired workers are separately identified in the LEHD data, allowing me to calculate the average earnings of only newly hired full-time workers (versus all full-time workers) in each plant-year. This creates a new outcome variable, one which directly speaks to the issue of downward wage rigidities.

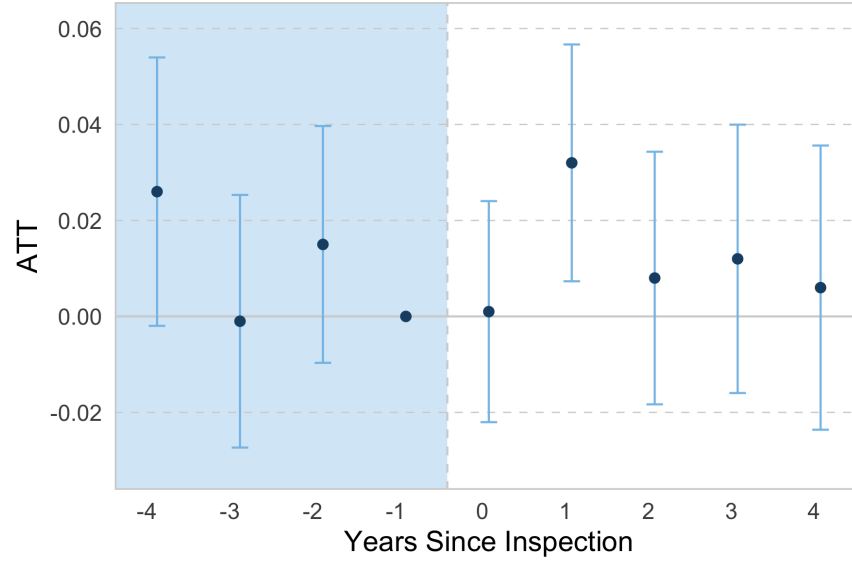
Even if downward wage rigidity prevents the earnings of the average full time worker from changing in response to OSHA enforcement, for newly hired workers there is no precedent wage from which to decrease. This should weaken or negate the effect of the frictions which drive the Shimer (2005) hypothesis, as firms would have greater license or freedom to set lower wages for new hires given they are only ever exposed to the new safer conditions.

I find that OSHA enforcement does not significantly affect the earnings of newly hired workers in the full sample, as shown below in Figure 13. If anything, there is a slight increase in earnings in the first full year after the OSHA inspection takes place, of roughly 3%. Unlike earnings for all workers, the earnings of new-hires does display heterogeneity by sub-sample. Low fines display a null effect (similar to the result for all workers), while high fines surprisingly seem to have a positive effect. This is confirmed in the difference-in-difference analysis in Table 6, where an estimated effect of a 4.5% increase in earnings is estimated.

The result for high fines is particularly interesting, as it is the opposite of what a compensating differentials theory would predict: safer workers are now paid more than their counterparts who did not experience an increase in safety. This contrasts not only with the prediction of compensating differentials theory, but also with the rigid wages literature. Studies of specifically newly-hired workers tend to find that they are paid quite similarly to existing workers, out of firms desire for fairness and to avoid discouraging effort (Galuscak et al. 2012). That they are paid more in the case of high regulatory fines suggests that there

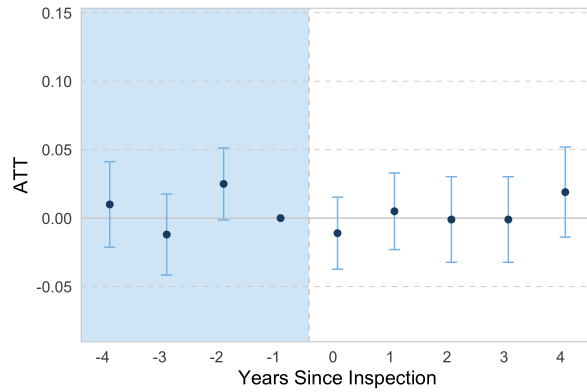
is another force countervailing these frictions.

Figure 13: Event Study ATT Estimates - Log Average Quarterly New-Hire Earnings



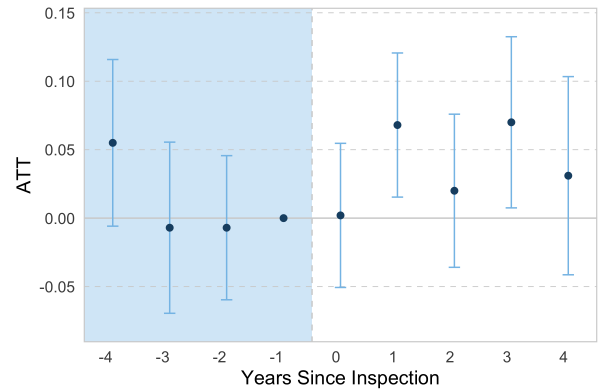
Note: Event study coefficients from Equation (1) for log quarterly new-hire earnings to the inspection year (year 0), with 95% confidence intervals. The year before inspection is omitted (year -1). The shaded area represents the pre-inspection period.

Figure 14: ATTs - New-Hire Earnings, Low Fine



Note: Event study coefficients Equation (1) for log quarterly new-hire earnings in the low-fine sample with 95% confidence intervals. The year before inspection is omitted (year -1).

Figure 15: ATTs - New-Hire Earnings, High Fine



Note: Event study coefficients Equation (1) for log quarterly new-hire earnings in the high-fine sample with 95% confidence intervals. The year before inspection is omitted (year -1).

Table 6: Difference-in-Difference Effects on New-Hire Earnings

	Log Average Quarterly New-Hire Earnings		
	(1)	(2)	(3)
	All Fines	Low Fine	High Fine
Post $\times$ Fine	0.015 (0.009) [0.116]	0.004 (0.011) [0.684]	0.043** (0.020) [0.030]
Plant FE	✓	✓	✓
State-Year FE	✓	✓	✓
Industry-Year FE	✓	✓	✓
R^2	0.902	0.912	0.885
N	53,500	37,000	16,500

Note: Standard errors are clustered at the matched-pair level in parentheses; P-levels in brackets.

* = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

This result suggests an additional indirect cost to regulatory enforcement, as discussed in [Section V.A.2](#). Enforcement of high fines may be causing reputational damage to the plant, identifying it as an unsafe workplace. It may also be a signal of a poorly managed plant, following the findings of [Bloom et al. \(2025\)](#). In this way, OSHA enforcement of particularly high fines may act as an information shock to potential new hires, causing them to demand higher wages as a result. This aligns with the effects found in [Viscusi and O'Connor \(1984\)](#) using experimental information treatments about safety. Workers could find out about OSHA enforcement through a variety of mechanisms, including press releases, media coverage, or word of mouth in particular industries.

This finding identifies a concrete indirect cost for firms experiencing high fines: higher labor costs for new hires. This additional cost supports the profitability-based interpretation of the results; however, this support is contingent upon the assumption that the higher wages are not reflective of efficient payment for higher talent / ability. If high-fined plants are paying more for new hires, it could be the case that they are hiring more talented workers and getting more productivity out of them, which would negate the effect on profitability.

To test this assumption, I estimate a model which identifies the average worker talent through fixed effects analysis ([Abowd et al. 1999](#)). The results are reported in full in [Appendix E](#), but do not support the assertion that high-fine plants are gaining more talented workers after inspection. If anything, the results suggest a negative overall effect on worker talent across fined plants.

VII Conclusion

Opponents of business regulation argue it stifles economic activity, depressing employment and hurting businesses. While this may hold in some contexts ([Karpoff et al. 2005](#); [Ryan 2012](#); [Curtis 2020](#)), I provide new evidence that the micro-level effects of regulatory enforcement are more complicated. I find that the effects of enforcement depend heavily on the balance between direct and indirect costs and benefits of remediation.

While enforcement resulting in relatively high fines can depress employment and make plants more likely to close, in contrast when relatively low fines are assessed the opposite is true: plants increase employment and are more likely to remain open. While the direct effects of enforcement on the costs and benefits of a plant do not explain these results, consideration of the indirect effects as well provides a more consistent interpretation. This emphasizes the importance of costs like productivity losses and reputational damage, and of benefits like spillover improvements outside of safety.

These findings could inform improvements in regulatory design. OSHA inspectors could more proactively communicate about the opportunity for spillover improvements, building on their experience across plants which have previously remediated similar hazards successfully. They could also help mitigate indirect costs, by emphasizing safety improvements when they occur, potentially reducing the reputational damage of high fines. More generally, these results suggest that increased enforcement of low-fine violations could have positive effects.

This analysis holds for safety regulation, but may not be applicable to other areas of regulation and regulators. Safety remediation has relatively clear implications in cost-saving for the firm, whereas environmental or workers-rights regulation does not. Additionally, the effects of OSHA enforcement may be driven by agency-specific nuances; future research into cross-regulator differences would help understand the external validity of these results.

Regulatory enforcement does not have uniform effects. They depend on what standards are being enforced, the associated penalties, and the indirect costs and benefits which come from remediation. This finding encourages future research into the multidimensional effects of regulatory enforcement, to better understand the indirect impacts on plants. Additionally, studies which can estimate the long-term impacts on workers would provide a more holistic understanding of the welfare effects of regulatory enforcement.

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Appendix

A Model Extension to Uncertainty

Further insights could be achieved by introducing two additional elements to the model: uncertainty over inspections and fines for unremediated hazards. These are realistic assumptions: OSHA inspects only one percent of all plants in its jurisdiction each year, and considers a diverse array of factors when selecting plants for inspection (reviewed in [Section III](#)) — OSHA also levies hundreds of millions of dollars in fines each year.

Extending the model this way, plants make their *ex ante* safety choices given a belief on the probability of inspection, $\pi \in [0, 1]$. Combining this with their belief on the costs and benefits of remediation, the decision for each hazard now depends on comparison of two options:

1. Remediate the hazard, incurring (in their belief) $\tilde{C}_h$ and gaining $\tilde{\delta}_h$ with certainty
2. Leave the hazard unremediated, with two possible outcomes:
 - With probability $1 - \pi$, there is no inspection and profitability is unchanged
 - With probability π , there is an inspection, the firm must remediate (incur $\tilde{C}_h$ and gain $\tilde{\delta}_h$), and further must pay an associated fine F_h

Plants will therefore remediate if the following inequality holds:

$$\tilde{\delta}_h - \tilde{C}_h \geq \pi(\tilde{\delta}_h - \tilde{C}_h - F_h)$$

Rearranging this inequality, the role of each factor is intuitive. The value of remediation is increasing in the perceived benefit $\tilde{\delta}_h$ which enters positively, but decreasing in the cost $\tilde{C}_h$ which enters negatively. The value of leaving the hazard is decreasing in the fine F_h , as well as the probability of inspection.

$$\tilde{\delta}_h - \tilde{C}_h \geq -F_h \frac{\pi}{1 - \pi}$$

Under perfect information, this extension would introduce no heterogeneity of results, as plants would still always remediate profitable hazards and not others, so that all inspections would result in enforcement of remediating unprofitable hazards. However, under imperfect information, this modification adds to the degree of variation possible, amplifying the magnitude of incorrect beliefs through beliefs over probability of inspection and fine values.

Additional heterogeneity in outcomes could be introduced through additional extensions on the plant side, such as allowing beliefs to vary (π_j for each plant j) or introducing more complex utility functions such that some plants are more risk averse or risk prone.

B Data Linkages

This section describes the two major data linkages which were required to conduct the empirical analysis of this project. The first linkage was between the public OSHA inspection archive and the LBD at the US Census. The second was from the LBD to the LEHD within the Census.

The LBD is often linked to data external to the Census via an intermediary dataset, the County Business Patterns Business Register (CBPBR). This dataset operates similarly to the LBD, including information from every business in every year, but crucially has the additional variables of name and address (whereas the LBD only includes anonymized identifier variables, such as LBD Number). Individual plant-year observations are cleanly linked one-to-one between the LBD and CBPBR using common administrative variables in both, called `id` in the CBPBR and `estabid` in the LBD.

Once the LBD is linked to the CBPBR, the names and addresses of each plant can be used in a fuzzy-linking methodology to link outwards to the OSHA data. Helpfully, both datasets contain not one but two name variables, as is common practice for businesses. Usually, these reflect one “trade name” and one more formal “doing-business-as” name (for example, “Restaurant Holdings LLC” and “John’s Restaurant”). The following procedure was used:

1. Block data into cells of observation year, state, county, and industry in both datasets
2. Within each cell, conduct fuzzy matching of observations between the datasets on address and name variables, using the `reclink2` command in Stata (conducting a separate matching for each of the four possible name variable permutations)
3. Refine the identified viable matches to restrict to the one OSHA observation with the highest matching score (produced by `reclink2` for string similarity) for each LBD observation
4. Use simple string processing rules to eliminate matches which are clearly poor quality
5. Spot check the remaining matches for one-off corrections

The resulting dataset matched approximately 40% of the OSHA archive in to the LBD, on par with similar efforts in related projects ([Bloom et al. 2025](#)).

The second linking was internal to the Census, between the LBD and the LEHD QWI dataset. Unlike the LBD and CBPBR, there is no single common identifying variable between these two datasets at the plant-year observation level. Instead, there is a variable which is common to both and identifies unique firms in each year, called `EIN` (for Employer Identification Number) in the LBD and `SEIN` in the LEHD (State EIN). While it is possible

to match observations at the plant-year level one-to-one using this variable, that is only true for those plants which are the sole operating plant of the broader firm in a given state. In other words, Firm A can have one plant in Maine and two plants in Massachusetts. Using this method, it would be possible to link the plant in Maine, as only one plant would appear in Maine with Firm A's EIN in both datasets. For the two plants in Massachusetts, both appear with the same EIN in both datasets, making the matching of which-is-which more difficult.

To overcome this difficulty, I leverage additional variables which are similar in both datasets to try and refine the matching. Specifically, I use the industry codes and detailed geographic variables to restrict the number of possible matches to one in each case. For example, if the two plants in Massachusetts above are in different counties, I would then match between the two datasets on year-EIN-county, and find only one possible match in each case. Alternatively, if the two plants are in the same county but one is a manufacturing facility while the other is a logistics trade hub, they would have different industry codes, but the same codes in each dataset. I iterate over several combinations of such identifying variables to identify all possible unambiguous one-to-one matches between the two datasets.

C Balance Table

The balance between treatment and control samples is also balanced within the two sub-samples defined by the low versus high fine treatments.

Table 7: Summary Statistics

	Low Fine		High Fine	
	Control	Treated	Control	Treated
Firm Age	20.9 (11.090)	21.0 (11.000)	20.1 (12.290)	18.2 (12.360)
Log Plant Size	3.32 (1.512)	3.60 (1.305)	2.79 (1.441)	2.16 (1.131)
Log Firm Size	4.50 (2.877)	4.61 (2.601)	3.58 (2.713)	2.71 (2.119)
Log Quarterly Worker Earnings	8.01 (0.553)	7.98 (0.480)	8.05 (0.604)	7.94 (0.584)
Log Payroll (000s)	6.84 (1.692)	7.10 (1.455)	6.36 (1.692)	5.61 (1.405)
Log Quarterly New Hire Earnings	7.61 (0.682)	7.58 (0.611)	7.68 (0.695)	7.55 (0.716)
County-Sector Employment HHI	0.283 (0.699)	0.299 (0.678)	0.244 (0.668)	0.209 (0.484)
Plants	2,600	2,600	1,200	1,200

Note: Means are provided with standard deviations in parentheses.

D Estimating OSHA Standard Remediation Costs

I used a keyword-based text algorithm to estimate the costs of compliance with OSHA standards. This approach combines tractability and replicability with sufficient richness to generate variation.

I constructed a dataset of OSHA texts by scraping the Electronic Code of the Federal Register, using the standards enumerated in the OSHA inspections data. I scraped the specific text of the regulatory statute — for example, for standard 1926.1153(g)(4), I extract:

“The employer shall designate a competent person to make frequent and regular inspections of job sites, materials, and equipment to implement the written exposure control plan.”

I then studied the key words which appeared frequently across standards and developed a set of cost logic rules for the algorithm, displayed in [Table 8](#). Costs were based on an assumption of 10 employees.

Table 8: Cost Estimation Rules Based on Standard Keywords

Category	Keywords	Cost Logic
Training Costs	train, training program, ensure employee, inform employees	+\$1,500 +\$500 if annual
Medical Surveillance	medical, healthcare professional, physician, examination, medical surveillance	+\$3000; +\$1000 if annual or periodic
Respiratory Protection Program	respirat	+\$6,000
Exposure Monitoring	monitor, monitoring, exposure assessment, air sample, breathing zone	+\$4,000; +\$2,000 if initial and periodic; +\$3,000 if quarterly
Written Program Development	written program, written plan, exposure control plan, establish a written	+\$2,000; +\$500 if review and update
Personal Protective Equipment	protective clothing, protective equipment, ppe, gloves, masks, face shield, goggles, apron, gown, coverall	+\$1,000; +\$500 if provide and no cost
Engineering Controls	engineering control, ventilation, enclosure, exhaust, isolation, local exhaust	+\$20,000; +\$5,000 if hepa
Signage and Labeling	sign, label, posted, warning, caution, danger	+\$500
Record Keeping	record, maintain records, documentation, keep a copy	+\$1,200
Inspection Programs	inspect, inspection, qualified person	+\$2,000; +\$1,500 if monthly; +\$2,500 if daily or each shift
Specialized Equipment	ladder, scaffold, crane, hoist, rigging	+\$3,000
Hazard Communication Program	hazard communication, safety data sheet, sds	+\$3,000
Emergency Procedures	emergency, evacuation, rescue, emergency response	+\$3,500
Confined Space Program	permit space, confined space	+\$6,000
Blood	blood, pathogen, hepatitis, hiv, sharps	+\$5,000
Asbestos	asbestos, acm	+\$5,000
Toxic Chemicals	lead, silica, formaldehyde, cadmium, chromium	+\$5,500
Facilities	shower, change room, lunchroom, washing facilities	+\$10,000
Competent Person Designation	competent person	+\$5,000
Cleaning and Housekeeping	clean, housekeeping, decontamination, waste disposal	+\$2,000

E AKM Analysis

When estimating the effects of enforcement on worker earnings, analysis at the plant level suffers from a lack of information about the underlying workers. Without knowing the distribution of worker characteristics (particularly worker ability), the interpretation of plant-level regressions is limited. Changes could be efficient responses to changes in workers, or could be inefficient changes resulting from frictions.

For example, if wages increase following enforcement, this could reflect of increasing worker talent or ability leading to efficiently higher wages. Alternatively, it could be that the plant is forced to overpay relative to talent due to reputational damage from OSHA enforcement (with no change in worker ability). Without worker information, we cannot disentangle these two explanations.

I estimate a model of worker earnings following [Abowd et al. \(1999\)](#), a common way to overcome this problem. This model (henceforth AKM) incorporates both worker fixed effects and firm fixed effects to the earnings process. This allow me to then consider the estimated worker fixed effects as a proxy for worker ability, as it measures how much individuals earn relative to expectations based on their places of employment. I use an AKM model to estimate the average worker earning ability at each plant in each year, using the following two-step approach. In a first step, I estimate a simple AKM model using all workers, plants, and years available in the LEHD data:

$$\ln w_{it} = \omega_i + \pi_{j(it)} + \beta X_{it} + \epsilon_{it} \quad (3)$$

where w_{it} is the total earnings of worker i in quarter t ¹¹. This is modelled as a function of the worker’s own fixed effect, ω_i , the effect of the plant at which the worker is in that quarter, $\pi_{j(it)}$, and a vector of time varying individual control variables X_{it} , which is essentially age polynomials interacted with education following [Card et al. \(2013\)](#). Although it is possible in principle to have estimated this step using only the analysis sample, the smaller sample size would have meant significantly noisier estimates of the effects.

In a second step, I then estimate the average $\hat{\omega}_i$ of workers in set I at each plant j in

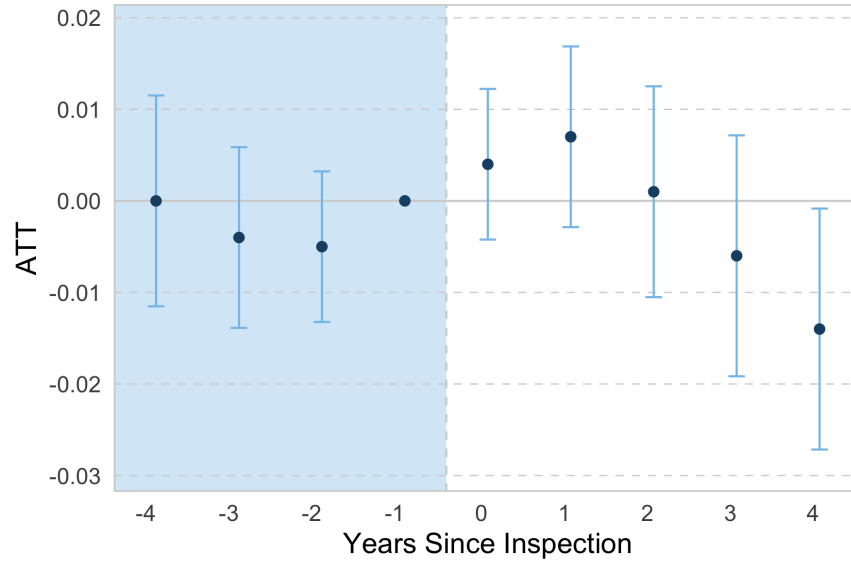
¹¹The LEHD identifies all jobs at which a worker is employed in every quarter, resulting in multiple earnings streams. I restrict the sample to include only one observation per worker-quarter, using that of the highest earnings value.

each year t as:

$$\bar{\omega}_{jt} = \frac{\sum_{i \in I_{jt}} \hat{\omega}_i}{I_{jt}} \quad (4)$$

I then use this $\bar{\omega}_{jt}$ as an outcome in the aforementioned event study and difference-in-difference specifications. The results presented in Figure 16 suggest that worker talent is not significantly impacted by OSHA enforcement. The point estimates suggest a possible immediate rise in average worker talent post enforcement, followed by a steady decline in the subsequent years, though the effects are small and all estimates are imprecise.

Figure 16: Event Study ATT Estimates - Average Worker Effect



Note: Event study coefficients from Equation (1) for average worker effects relative to the inspection year (year 0), with 95% confidence intervals. The year before inspection is omitted (year -1). The shaded area represents the pre-inspection period.

These results lack a clear implication for the relationship between safety and worker composition.